

Exact Inversion of the Partition Function

A proof of OEIS A306631 and sharp rounding bounds

Convergent inverse corrections, arithmetic exponential terms,
and exact finite verification

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Abstract

Let $p(n)$ be the ordinary partition function, and let F be the increasing inverse of $H(x) = e^{\pi\sqrt{2x/3}}/(4\sqrt{3}x)$. We prove the conjecture recorded in OEIS A306631 that rounding $F(p(n))$ recovers n for every $n > 9$. More precisely, its only failures for $n \geq 2$ occur at 2, 3, 4, 5, 7, 9, whereas taking the ceiling succeeds for every $n \geq 2$. The inverse bias $d_n = n - F(p(n))$ has the sharp, unattained infimum $\delta = 1/24 + 3/\pi^2$. Its maximum on $n \geq 2$ occurs uniquely at 2; its maximum on $n \geq 10$ occurs uniquely at 11, where the bias is about 0.4984875754. These extrema determine every constant shift that permits uniform exact rounding, the unique minimax shift, and a two-candidate inversion formula requiring one comparison with a partition value.

A convergent power series describes the smooth dominant-sector bias, with an explicit geometric remainder bound. A second convergent series inverts the corrected dominant model. We compute twelve coefficients of each, prove a finite hierarchy of arithmetic exponential corrections, identify the leading parity term, and deduce fixed-order eventual alternating finite differences. The proof combines Rademacher's classical formula with an elementary tail estimate for $n \geq 400$ and a reproducible rational-interval certificate for the remaining indices.

Status. This is an unrefereed research report. The analytic proofs and exact finite certificate are supplied, but are not formalized in Lean or Rocq. The literature and repository checks establish the specific conjecture being addressed, not publication priority for every consequence. Classical partition asymptotics and Rademacher's formula are not claimed as new.

Contents

1	The problem, its status, and the main results	3
1.1	A precise OEIS question	3
1.2	What is established here, and what is classical	4
1.3	Why ordinary asymptotic equivalence is insufficient	4
2	The inverse branch and the exact partition expansion	4
2.1	Branch selection	4
2.2	The external theorem and our normalization	5
3	An elementary effective Rademacher tail	5
4	A convergent model for the inverse bias	6
5	Uniform inequalities for every large index	7
6	The exact finite certificate and completion of the proof	8
6.1	What must be checked	8
6.2	Why the intervals are rigorous	9
6.3	Completion of the main theorem	10
7	Optimal constant shifts and robust rounding	10
8	Inverting the entire partition staircase	11
9	All-orders bias coefficients and an explicit remainder	12
10	A convergent corrected inverse at an arbitrary target	13
11	Arithmetic exponential corrections	14
12	Consequences for differences and perturbations	16
12.1	Fixed-order eventual alternating differences	16
12.2	The natural noise scale for recovering an index	17
12.3	Relation to the forward approximation in A050811	17
13	Reproducibility and numerical illustrations	18
13.1	Three deliberately separate forms of verification	18
13.2	The small indices that decide the theorem	18
13.3	Checking the parity term and the convergent tails	19
14	Further research questions and topics	20
15	Conclusion	21
A	A compact proof dependency ledger	22
B	Suggested OEIS update and source audit	22

1 The problem, its status, and the main results

1.1 A precise OEIS question

The OEIS entry A306631 defines the nearest-integer value of an inverse Hardy–Ramanujan approximation, expressed using the real branch W_{-1} of the Lambert function. In the entry inspected on 3 October 2026, the assertion that this sequence takes the value n at the argument $p(n)$ for every $n > 9$ is explicitly labeled a conjecture [1]. Thus the question concerns an *exact integer recovery property*, not merely asymptotic inversion.

The ordinary partition function $p(n)$, OEIS A000041 [2], counts unordered partitions of n into positive integers; $p(0) = 1$. Put

$$a = \frac{1}{24}, \quad c = \pi\sqrt{\frac{2}{3}}, \quad \delta = a + \frac{2}{c^2} = \frac{1}{24} + \frac{3}{\pi^2}. \quad (1.1)$$

For $x > 0$, define

$$H(x) = \frac{e^{c\sqrt{x}}}{4\sqrt{3}x}. \quad (1.2)$$

The function H is strictly increasing for $x > 4/c^2$. Its inverse on that branch, restricted to $y \geq 2$, is

$$F(y) = \frac{6}{\pi^2} W_{-1} \left(-\frac{\pi}{2\sqrt{2}3^{3/4}\sqrt{y}} \right)^2. \quad (1.3)$$

Write

$$d_n = n - F(p(n)), \quad n \geq 2, \quad \text{rnd}(x) = \lfloor x + \tfrac{1}{2} \rfloor. \quad (1.4)$$

The rounding convention at exact half-integers will matter for one endpoint of a shifted-rounding theorem, but not for the original conjecture: all its relevant comparisons are strict.

Theorem 1.1 (Exact rounding and sharp bias bounds). *For every integer $n \geq 2$,*

$$0 < d_n < 1, \quad \lceil F(p(n)) \rceil = n. \quad (1.5)$$

The complete nearest-integer classification is

$$\text{rnd}(F(p(n))) = \begin{cases} n-1, & n \in \{2, 3, 4, 5, 7, 9\}, \\ n, & \text{otherwise.} \end{cases} \quad (1.6)$$

Moreover, with

$$\beta_2 = 2 - F(2), \quad \beta_{11} = 11 - F(56), \quad (1.7)$$

one has the sharp inequalities

$$\delta < d_n \leq \beta_2 \quad (n \geq 2), \quad (1.8)$$

$$\delta < d_n \leq \beta_{11} \quad (n \geq 10). \quad (1.9)$$

Equality on the right holds only at $n = 2$ and $n = 11$, respectively. In both ranges the infimum is δ , is not attained, and is approached as $n \rightarrow \infty$.

The conjectured threshold 10 is therefore sharp: 9 really is a failure. More surprisingly, the index closest to failure *after* the threshold is 11, not 10. Certified enclosures give

$$\begin{aligned}\delta &= 0.34563021759367998099830505629584958\dots, \\ \beta_2 &= 0.79661548126519453324805686994201716\dots, \\ \beta_{11} &= 0.49848757539660686901336470177641810\dots\end{aligned}$$

Sections 3–6 prove the theorem. The reader interested only in resolving A306631 may stop after that proof.

1.2 What is established here, and what is classical

Rademacher’s convergent formula is the external analytic input; we use the normalization in Johansson [4]. Effective asymptotic expansions of the partition function itself have an extensive literature; Banerjee, Paule, Radu and Schneider [5] are a relevant modern reference. None of that machinery is presented here as a new discovery. The central deliverables of this report are the exact rounding classification, the sharp extrema and calibration statements, and explicitly controlled inverse formulas derived from that input.

The ProveIt repository already contains work on all-orders asymptotics, inverse errors and integer thresholds. In particular, the inspected A292507 report [6] treats the binomial transform of partitions without ones. The present problem continues that methodology but concerns ordinary partition values and a *global exact* rounding statement. The repository snapshot used for the source review was

d68b65ea064084fb121df9bb431b21d086f7be81.

A targeted repository search for A306631 returned no match. This is a limited search result, not a guarantee that the subject is absent from every repository file or from the literature. No repository theorem is assumed in the proofs below, and no repository file has been modified.

1.3 Why ordinary asymptotic equivalence is insufficient

From $p(n) \sim H(n)$ one obtains only a coarse inverse approximation. The derivative

$$(\log H)'(n) \sim \frac{c}{2\sqrt{n}}$$

tends to zero. A relative forward error of size $n^{-1/2}$ can therefore produce a nonzero constant inverse error. In fact the error tends to δ , rather than to zero. A proof that rounding succeeds must place the *entire* error sequence on the correct side of $1/2$; the exceptional behavior at small odd indices cannot be discarded.

2 The inverse branch and the exact partition expansion

2.1 Branch selection

Logarithmic differentiation gives

$$(\log H)'(x) = \frac{c}{2\sqrt{x}} - \frac{1}{x} = \frac{c^2(t-2)}{2t^2}, \quad t = c\sqrt{x}. \quad (2.1)$$

The minimum of H is $c^2 e^2 / (16\sqrt{3}) < 2$, so the increasing inverse exists for all $y \geq 2$. The elementary bounds $\pi^2 < 10$, $e < 11/4$ and $\sqrt{3} > 5/3$ already imply that this minimum is less than $121/64 < 2$. Solving $H(x) = y$ with $t = c\sqrt{x} > 2$ gives

$$-\frac{t}{2} e^{-t/2} = -\frac{c}{4 \cdot 3^{1/4} \sqrt{y}}.$$

The choice $t > 2$ selects W_{-1} and proves (1.3). The principal Lambert branch would instead select the decreasing branch and would not solve the OEIS question.

2.2 The external theorem and our normalization

For $n \geq 1$ write

$$m = n - a, \quad \mu = c\sqrt{m}, \quad J(t) = t \cosh t - \sinh t. \quad (2.2)$$

For relatively prime integers h, k , the Dedekind sum is

$$s(h, k) = \sum_{r=1}^{k-1} \left(\left(\frac{r}{k} \right) \right) \left(\left(\frac{hr}{k} \right) \right),$$

where $((u)) = u - \lfloor u \rfloor - 1/2$ for nonintegral u , and $((u)) = 0$ for integral u . Define

$$A_k(n) = \sum_{\substack{0 \leq h < k \\ (h, k) = 1}} \exp \left(\pi i \left(s(h, k) - \frac{2nh}{k} \right) \right). \quad (2.3)$$

For $k = 1$ the convention yields $A_1(n) = 1$; also $A_2(n) = (-1)^n$. Termwise absolute values give $|A_k(n)| \leq k$. Each $A_k(n)$ is periodic in n modulo k .

Rademacher's formula, in the equivalent normalization of [4, Section 1], is

$$p(n) = \frac{1}{2\pi\sqrt{2} m^{3/2}} \sum_{k \geq 1} A_k(n) \sqrt{k} J(\mu/k). \quad (2.4)$$

The classical identity (2.4) is quoted, not reproved. The estimates, inversions and finite verifications that follow are proved within this report.

3 An elementary effective Rademacher tail

Lemma 3.1 (Absolute tail bound). *For every $K \geq 1$ and $n \geq 1$, if $T_k(n)$ is the k th term of (2.4), then*

$$\left| p(n) - \sum_{k=1}^K T_k(n) \right| \leq \frac{2\pi^2}{9\sqrt{3}K} e^{\mu/(K+1)}. \quad (3.1)$$

Proof. For $t \geq 0$,

$$J(t) = \int_0^t u \sinh u \, du \leq \int_0^t u^2 e^u \, du \leq \frac{t^3 e^t}{3}.$$

Consequently

$$|T_k(n)| \leq \frac{c^3}{6\pi\sqrt{2}} k^{-3/2} e^{\mu/k} = \frac{\pi^2}{9\sqrt{3}} k^{-3/2} e^{\mu/k}.$$

This also proves absolute convergence for fixed n . For $k > K$ use $e^{\mu/k} \leq e^{\mu/(K+1)}$ and $\sum_{k>K} k^{-3/2} \leq \int_K^\infty x^{-3/2} dx = 2/\sqrt{K}$. $\square$

The bound is deliberately elementary rather than optimal. Its purpose is to turn an infinite-index claim into a finite rational certificate, not to compete with optimized implementations of Rademacher's formula.

Define the positive dominant sector, extended to real $x > a + c^{-2}$, by

$$M(x) = \frac{e^{c\sqrt{x-a}}}{4\sqrt{3}(x-a)} \left(1 - \frac{1}{c\sqrt{x-a}}\right). \quad (3.2)$$

Splitting the $k = 1$ summand into its positive and negative exponentials shows that

$$T_1(n) = M(n) + \frac{e^{-\mu}}{4\sqrt{3}m}(1 + \mu^{-1}).$$

Lemma 3.2 (Relative error of the dominant sector). *For $\mu \geq 50$, write $p(n) = M(n)(1 + \epsilon_n)$. Then*

$$|\epsilon_n| \leq \frac{3}{2}\mu^2 e^{-\mu/2}. \quad (3.3)$$

Proof. The $K = 1$ case of (3.1) and the displayed negative exponential term imply

$$|\epsilon_n| \leq \frac{1 + 1/\mu}{1 - 1/\mu} e^{-2\mu} + \frac{4}{3} \frac{\mu^2}{1 - 1/\mu} e^{-\mu/2}. \quad (3.4)$$

For $\mu \geq 50$, the coefficient of $\mu^2 e^{-\mu/2}$ in the second term is at most $200/147$. The first term divided by that same scale is at most $51/122500$, even after discarding its additional decaying exponential. The sum of those rational bounds is less than $3/2$. $\square$

4 A convergent model for the inverse bias

The dominant sector can be inverted without making a merely formal asymptotic ansatz. Its bias is an analytic function of $1/\mu$.

Lemma 4.1 (Analytic fixed point). *There is a unique analytic solution with $|v(z) - 1| \leq 1/4$ on $|z| \leq 1/20$ of*

$$v(z) = -\frac{\log(1-z)}{z} - \frac{2}{z} \log(1 - z^2 v(z)), \quad v(0) = 1. \quad (4.1)$$

The apparent singularities at $z = 0$ are removable. All Taylor coefficients of v are nonnegative. In particular, for $0 \leq z \leq 1/20$,

$$1 + \frac{5}{2}z \leq v(z) \leq \frac{5}{4}. \quad (4.2)$$

Proof. Use the analytic logarithms determined by their series at 1. Put $r = 1/20$ and $Q = 5/4$. On $|z| \leq r$, $|v - 1| \leq 1/4$, the right-hand side of (4.1), minus 1, has modulus at most

$$\frac{r}{2(1-r)} + \frac{2rQ}{1-r^2Q} < \frac{1}{4}.$$

Its derivative with respect to v has modulus at most

$$\frac{2r}{1-r^2Q} < \frac{1}{4}.$$

Uniform fixed-point iteration starting at $v = 1$ is thus a contraction on that disk, and converges uniformly to an analytic solution. The same bounds hold in a slightly larger z -disk, so analyticity on a neighborhood of the closed disk is available for Cauchy estimates.

Expanding (4.1) gives

$$v_0 = 1, \quad v_j = \frac{1}{j+1} + 2 \sum_{k=1}^{\lfloor (j+1)/2 \rfloor} \frac{1}{k} [z^{j-2k+1}] v(z)^k \quad (j \geq 1). \quad (4.3)$$

Only earlier coefficients occur on the right, and they occur with nonnegative coefficients. Induction proves $v_j \geq 0$, while $v_1 = 5/2$. The upper real bound is the fixed-point disk bound. $\square$

Definition 4.2. For $|z| \leq 1/20$, set

$$B(z) = a + \frac{2v(z) - z^2 v(z)^2}{c^2}. \quad (4.4)$$

Proposition 4.3 (Exact dominant-sector identity). *Whenever $\mu = c\sqrt{n-a} \geq 50$,*

$$n - F(M(n)) = B(1/\mu). \quad (4.5)$$

The same identity holds for real n in this range.

Proof. Set $z = 1/\mu$ and $\nu = \mu - zv(z)$. Equation (4.1) is exactly

$$-zv - 2 \log(1 - z^2 v) = \log(1 - z).$$

Hence

$$\frac{e^\nu}{\nu^2} = \frac{e^\mu}{\mu^2} (1 - \mu^{-1}).$$

Since $\nu > 2$, the increasing inverse branch gives $F(M(n)) = \nu^2/c^2$. Subtracting this quantity from $n = a + \mu^2/c^2$ proves the assertion. $\square$

Remark 4.4. The exact identity concerns $M(n)$, not $p(n)$. The distinction is essential: $B(1/\mu)$ is a convergent analytic main sector, whereas d_n also contains arithmetic exponentially small terms. These terms will be made explicit in Section 11.

5 Uniform inequalities for every large index

Theorem 5.1 (Effective inverse estimate). *For every integer $n \geq 400$, with $\mu = c\sqrt{n-a}$,*

$$|d_n - B(1/\mu)| \leq \frac{9\mu^3}{c^2} e^{-\mu/2} < \frac{1}{c^2 \mu}. \quad (5.1)$$

In particular,

$$\delta + \frac{3}{c^2 \mu} < d_n < \frac{43}{100}, \quad \lim_{n \rightarrow \infty} d_n = \delta. \quad (5.2)$$

Proof. The elementary bounds $25/8 < \pi < 22/7$ imply $625/96 < c^2 < 7$. Thus $n \geq 400$ gives $\mu^2 > (625/96)(400 - 1/24) > 2500$, so $\mu \geq 50$. Let $z = 1/\mu$. From (4.2),

$$B(z) - \delta \geq \frac{5z - (25/16)z^2}{c^2}, \quad 0 < B(z) \leq a + \frac{5}{2c^2} < \frac{43}{100}. \quad (5.3)$$

Write $x_0 = F(M(n)) = n - B(z)$. We first locate $F(p(n))$ in $[n - 1, n + 1]$ rather than presupposing that location.

For $x \in [n - 1, n + 1]$ put $t = c\sqrt{x}$. Since $c^2 < 7$ and $\mu \geq 50$, we have $\mu - 1 < t < \mu + 1$. The function $(t - 2)/t^2$ decreases for $t > 4$. Equation (2.1) therefore gives

$$(\log H)'(x) \geq \frac{c^2(\mu - 1)}{2(\mu + 1)^2} \geq \frac{c^2}{3\mu}. \quad (5.4)$$

The last inequality is equivalent to $\mu^2 - 7\mu - 2 \geq 0$.

The function $\mu^4 e^{-\mu/2}$ decreases for $\mu \geq 50$, and

$$e^{25} > 9 \cdot 50^4. \quad (5.5)$$

In fact $e^{25} > 25^{11}/11! > 9 \cdot 50^4$, a comparison of integers after multiplication by $11!$. The accompanying exact certificate also checks the inequality. Consequently $9\mu^4 e^{-\mu/2} < 1$ throughout the range. Lemma 3.2 now gives $|\epsilon_n| < 1/2$ and

$$|\log(1 + \epsilon_n)| \leq 2|\epsilon_n| \leq 3\mu^2 e^{-\mu/2} < \frac{1}{3\mu^2}. \quad (5.6)$$

The point x_0 is more than $1/2$ from either endpoint of $[n - 1, n + 1]$, by (5.3). Integrating (5.4) from x_0 to those endpoints shows that changing $\log M(n)$ by the amount in (5.6) cannot cross either endpoint value of $\log H$; indeed $1/(3\mu^2) < c^2/(6\mu)$. Thus $F(p(n))$ lies inside that interval. Applying the mean value theorem there proves

$$|F(p(n)) - F(M(n))| \leq \frac{3\mu}{c^2} |\log(1 + \epsilon_n)| \leq \frac{9\mu^3}{c^2} e^{-\mu/2}.$$

Together with (5.5), this is (5.1).

Subtracting z/c^2 from the lower bound in (5.3) gives

$$d_n - \delta > \frac{4z - (25/16)z^2}{c^2} > \frac{3z}{c^2}.$$

For the upper bound use $z \leq 1/50$ to obtain

$$d_n < a + \frac{5/2 + 1/50}{c^2} = a + \frac{63}{25c^2} < \frac{43}{100}.$$

For the last inequality, $c^2 > 625/96$ gives $a + 63/(25c^2) < 160777/375000 < 43/100$. The same elementary bound gives $\delta < 5233/15000 < 7/20$. These are also checked in the certificate. Finally, $B(0) = \delta$ and the error in (5.1) tends to zero. $\square$

6 The exact finite certificate and completion of the proof

6.1 What must be checked

Every comparison below takes place at $x \geq 1$, where H is strictly increasing. Thus

$$d_n > s \iff p(n) < H(n - s), \quad d_n < s \iff p(n) > H(n - s). \quad (6.1)$$

This avoids evaluating a Lambert function in any proof check.

Lemma 6.1 (Certified finite inequalities). *The following strict inequalities hold:*

$$\frac{7}{20} < d_n < 1 \quad (2 \leq n \leq 399), \quad (6.2)$$

$$d_n < \frac{49}{100} \quad (10 \leq n \leq 399, n \neq 11), \quad (6.3)$$

$$\frac{498}{1000} < d_{11} < \frac{499}{1000}, \quad \frac{796}{1000} < d_2 < \frac{797}{1000}, \quad (6.4)$$

$$d_n < \frac{79}{100} \quad (3 \leq n \leq 9). \quad (6.5)$$

For $2 \leq n \leq 9$, the inequality $d_n > 1/2$ holds exactly at $n = 2, 3, 4, 5, 7, 9$; at $6, 8$ one has $d_n < 1/2$.

Proof by exact finite computation. The file `code/certify.py` implements (6.1) using only integer arithmetic, integer square roots and rational endpoints. It completed all 1,204 asserted comparisons, with every interval separated strictly from the comparison value. The independent recomputation of all partition values also agreed. The construction of the intervals, including their rigorous transcendental remainders, is given below so that the finite computation can be audited without trusting a floating-point library. Its machine-readable outcome is `results/certificate.json`. $\square$

6.2 Why the intervals are rigorous

All stored interval endpoints are integers divided by $S = 10^{70}$. Addition is exact at this scale; multiplication and division round lower endpoints downward and upper endpoints upward. Square roots use integer square roots of scaled nonnegative integers, with an extra unit on an upper endpoint unless the square is exact. These operations enclose the corresponding real operations.

For π , the program uses Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

Each arctangent is enclosed between its alternating-series partial sum and that sum plus the next term. The program uses 60 terms and 15 terms, respectively, before rounding the rational endpoints outward. The identity follows by the tangent addition formula, with the relevant angles in the first quadrant.

For $0 \leq t \leq 64$, it computes $\exp(t/64)$ through degree 80 and then squares the interval six times. On $0 \leq u \leq 1$ the omitted tail is bounded by

$$\sum_{j=81}^{\infty} \frac{u^j}{j!} \leq \frac{1}{81!} \sum_{r \geq 0} 82^{-r} = \frac{82}{81 \cdot 81!} < \frac{2}{81!} < \frac{1}{S}.$$

Adding one scaled unit to the upper endpoint is therefore sufficient before the six squarings. Every exponential argument required by the finite comparisons is below 64. The computation of H then uses only these operations on rational arguments.

Partition numbers are first computed by Euler's recurrence

$$p(n) = \sum_{j \geq 1} (-1)^{j-1} \left(p\left(n - \frac{j(3j-1)}{2}\right) + p\left(n - \frac{j(3j+1)}{2}\right) \right), \quad (6.6)$$

with $p(0) = 1$ and $p(t) = 0$ for $t < 0$. As an independent check, a second program loop multiplies the truncated factors $(1 - q^j)^{-1}$ through q^{399} . Equivalently, starting with $b_0 = 1$ and $b_n = 0$ for

$n > 0$, for each part size $j = 1, \dots, 399$ it performs $b_n \leftarrow b_n + b_{n-j}$ in increasing order of $n \geq j$. This enumerates the possible multiplicities of each part size. All 400 coefficients agree with (6.6). Thus even the pentagonal identity is not needed as an unchecked source of the finite data.

The asserted H comparisons are grouped as follows.

Claim checked	Comparisons
$7/20 < d_n < 1$ for $2 \leq n \leq 399$	796
$d_n < 49/100$ for $10 \leq n \leq 399$, $n \neq 11$	389
Two-sided brackets at $n = 2$ and $n = 11$	4
$d_n < 79/100$ for $3 \leq n \leq 9$	7
Comparison with $1/2$ for $2 \leq n \leq 9$	8
Total asserted H comparisons	1,204

The smallest certified absolute separation among these comparisons exceeds 0.0002600145755; it occurs in the check $d_2 < 0.797$. Additional exact checks establish $\delta < 7/20$, (5.5), and the upper bound at the end of Theorem 5.1. Rational bisection of H , again without Lambert W , supplies the displayed high-precision enclosures for β_2 and β_{11} .

6.3 Completion of the main theorem

Proof of Theorem 1.1. For $n \geq 400$, Theorem 5.1 gives $\delta < d_n < 0.43$. For $2 \leq n \leq 399$, Lemma 6.1 gives $0.35 < d_n < 1$, and $\delta < 0.35$. This proves $0 < d_n < 1$ and hence the ceiling identity.

For $n \geq 10$, the finite bound is below 0.49 except at 11, where it is below 0.499; the infinite tail is below 0.43. Thus $d_n < 1/2$ for all $n \geq 10$. The last assertion of Lemma 6.1 classifies the indices from 2 to 9. Since $0 < d_n < 1$, their rounded values are exactly those in (1.6).

The value $d_{11} > 0.498$ is larger than every other d_n on $n \geq 10$: those with $n < 400$ are below 0.49, and the rest are below 0.43. Similarly, $d_2 > 0.796$ exceeds the bounds 0.79 for $3 \leq n \leq 9$ and 0.499 for $n \geq 10$. Thus both maxima are unique. All d_n exceed δ , while Theorem 5.1 gives $d_n \rightarrow \delta$. This proves the sharp, unattained infima. $\square$

Remark 6.2 (Scope of the certificate). This is an exact-arithmetic computer-assisted proof of a finite lemma, not a numerical experiment at selected large indices. The infinite range has a separate analytic proof. Nevertheless, executing Python and checking its source is not the same as a small-kernel formal proof; no Lean or Rocq certification is claimed. Reproduction should use ordinary Python execution, not the `-O` flag, which removes assertions.

7 Optimal constant shifts and robust rounding

The sharp extrema provide more information than the truth of the original conjecture.

Theorem 7.1 (All successful shifts and the minimax shift). *Let S be either $\{n \in \mathbb{N} : n \geq 2\}$ or $\{n \in \mathbb{N} : n \geq 10\}$, and let β be β_2 or β_{11} , respectively. For a real constant s , the identity*

$$\text{rnd}(F(p(n)) + s) = n \quad (n \in S) \quad (7.1)$$

holds if and only if

$$\beta - \frac{1}{2} \leq s \leq \delta + \frac{1}{2}. \quad (7.2)$$

The unique constant shift minimizing the worst absolute real error is

$$s_* = \frac{\beta + \delta}{2}, \quad \inf_{s \in \mathbb{R}} \sup_{n \in S} |F(p(n)) + s - n| = \frac{\beta - \delta}{2}. \quad (7.3)$$

Proof. By the rounding convention, (7.1) is equivalent to $-1/2 \leq s - d_n < 1/2$ for every $n \in S$. The left inequalities require $s \geq \sup d_n - 1/2 = \beta - 1/2$. The right inequalities allow precisely $s \leq \delta + 1/2$: equality is allowed because $d_n > \delta$ for every finite n , while any larger s fails along a sequence tending to the infimum. This proves (7.2).

Since the range has supremum β and infimum δ ,

$$\sup_{n \in S} |s - d_n| = \max\{|s - \delta|, |s - \beta|\}.$$

The unique minimizer is the midpoint, with half the interval length as its value. $\square$

For $n \geq 10$, exact interval bisection gives

$$\begin{aligned} s_* &= 0.42205889649514342500583487903613384\dots, \\ E_* &= 0.07642867890146344400752982274028426\dots \end{aligned}$$

Thus a fixed shift reduces the worst real error on the conjectured range from almost $1/2$ to less than 0.077 . The set of all successful shifts is approximately

$$[-0.00151242460339313\dots, 0.84563021759367998\dots] \quad (n \geq 10).$$

For all $n \geq 2$, the lower endpoint instead becomes $0.29661548126519453\dots$. In particular $s = 1/2$ works for every $n \geq 2$, which is another expression of the ceiling theorem.

The smallest symmetric numerical-error margin of the unshifted formula on $n \geq 10$ is

$$\rho = \frac{1}{2} - \beta_{11} = 0.00151242460339313098663529822358189\dots \quad (7.4)$$

An evaluation error of absolute value less than ρ cannot change any of those rounded answers. No larger uniform symmetric margin can be valid, because the lower rounding boundary at $n = 11$ is exactly ρ away. At the boundary itself the result depends on the tie convention; our lower endpoint in (7.2) uses rounding half-integers upward.

8 Inverting the entire partition staircase

Exact inversion at partition values is different from inversion at an arbitrary real target. Define

$$N(y) = \min\{n \geq 0 : p(n) \geq y\}, \quad y \geq 42. \quad (8.1)$$

Here the minimum exists because $p(n)$ is unbounded, and the relevant partition values are strictly increasing. Adding a part 1 injects partitions of n into those of $n + 1$, while the one-part partition of $n + 1$ is missing from that image for $n \geq 1$.

Theorem 8.1 (Two candidates and one partition comparison). *For $y \geq 42$, put*

$$k(y) = \lfloor F(y) + \delta \rfloor + 1. \quad (8.2)$$

Then

$$N(y) = \begin{cases} k(y), & p(k(y)) \geq y, \\ k(y) + 1, & p(k(y)) < y. \end{cases} \quad (8.3)$$

Proof. Suppose first that $y > 42$, and set $n = N(y) \geq 11$. The inequalities $p(n-1) < y \leq p(n)$ imply

$$n - 1 - d_{n-1} < F(y) \leq n - d_n.$$

Using $\delta < d_j \leq \beta_{11}$ for $j \geq 10$ gives

$$F(y) + \delta < n < F(y) + 1 + \beta_{11}. \quad (8.4)$$

The interval length is $1 + \beta_{11} - \delta < 2$. Its smallest possible integer is $k = \lfloor F(y) + \delta \rfloor + 1$, so $n \in \{k, k+1\}$. Monotonicity of p distinguishes these two possibilities by the one comparison in (8.3). For $y = 42$, the finite bounds on d_{10} give $9 < F(42) + \delta < 10$, so $k = 10 = N(42)$ directly. $\square$

This counts evaluations of the *partition function*, assuming the integer k has been computed correctly. It is not a bit-complexity claim about Lambert W , nor a claim that a fixed floating-point precision always determines a floor for arbitrary real input. Certified intervals for $F(y) + \delta$ must resolve the floor, and an input exactly on a floor boundary needs an exact representation or an appropriate equality convention. The numerical experiment shipped here checks all 4,959 integer targets $42 \leq y \leq 5000$; the theorem itself covers all real $y \geq 42$.

9 All-orders bias coefficients and an explicit remainder

The fixed-point equation gives an exact rational algorithm for every coefficient of the analytic bias, not only its first few terms.

Theorem 9.1 (Convergent bias series). *On $|z| < 1/20$,*

$$B(z) = \delta + \frac{1}{c^2} \sum_{j \geq 1} w_j z^j, \quad (9.1)$$

where the v_j are given by (4.3) and

$$w_j = 2v_j - [z^{j-2}]v(z)^2 \quad (j \geq 1), \quad (9.2)$$

with a negative-index coefficient interpreted as zero. For every $J \geq 0$ and real $\mu > 20$,

$$\left| B(1/\mu) - \delta - \frac{1}{c^2} \sum_{j=1}^J w_j \mu^{-j} \right| \leq \frac{129}{256c^2} \frac{(20/\mu)^{J+1}}{1 - 20/\mu}. \quad (9.3)$$

Consequently, for $n \geq 400$, the difference between d_n and the same truncated series is at most the right-hand side of (9.3) plus $9\mu^3 e^{-\mu/2}/c^2$.

Proof. The series follows from Lemma 4.1 and (4.4). On $|z| = 1/20$, that lemma gives

$$|c^2(B(z) - \delta)| = |2(v-1) - z^2 v^2| \leq \frac{1}{2} + \frac{1}{400} \frac{25}{16} = \frac{129}{256}.$$

Cauchy's coefficient estimate gives $|w_j| \leq (129/256)20^j$. Summing the resulting geometric tail proves (9.3). The assertion for d_n follows from Theorem 5.1. $\square$

The first terms are

$$d_n = \delta + \frac{5}{c^2\mu} + \frac{29}{3c^2\mu^2} + \frac{113}{6c^2\mu^3} + \frac{823}{20c^2\mu^4} + \mathcal{O}(\mu^{-5}). \quad (9.4)$$

More precisely, the complete convergent analytic series is accompanied by the exponentially small remainder in (5.1). In powers of $n^{-1/2}$ the first terms become

$$d_n = \delta + \frac{5}{c^3}n^{-1/2} + \frac{29}{3c^4}n^{-1} + \left(\frac{5a}{2c^3} + \frac{113}{6c^5}\right)n^{-3/2} + \mathcal{O}(n^{-2}). \quad (9.5)$$

The shift $a = 1/24$ must not be dropped when converting higher terms.

The finite algorithm in `code/series.py` uses exact rational arithmetic. It is triangular: the coefficient v_j in (4.3) depends only on $v_0, \dots, v_{j-1}$, so any prescribed finite order requires only finitely many additions and multiplications. Positivity of v_j does *not*, without a further argument, establish positivity of all w_j , because (9.2) includes a subtraction. The twelve positive values computed below are not used to assert that stronger statement.

10 A convergent corrected inverse at an arbitrary target

The preceding series describes the error at a known index n . For inversion from a target y , one wants a correction expressed in terms of $F(y)$ instead. These are different series.

For $\mu = c\sqrt{x-a} > 1$, differentiation of (3.2) yields

$$(\log M)'(x) = \frac{c^2(\mu^2 - 3\mu + 3)}{2\mu^2(\mu - 1)} > 0. \quad (10.1)$$

The numerator is $(\mu - 3/2)^2 + 3/4$. Since M increases from 0 to $+\infty$ on $x > a + c^{-2}$, it has an inverse G on $y > 0$.

Theorem 10.1 (Convergent correction to the Lambert inverse). *There is a unique analytic function q satisfying $q(0) = 1$, $|q(w) - 1| \leq 1/4$ on $|w| \leq 1/20$, and*

$$wq - 3\log(1 + w^2q) + \log(1 - w + w^2q) = 0. \quad (10.2)$$

For $u = F(y)$, $t = c\sqrt{u} > 20$, and $w = 1/t$,

$$G(y) = u + a + \frac{2q(w) + w^2q(w)^2}{c^2} = u + \delta + \frac{1}{c^2} \sum_{j \geq 1} r_j t^{-j}. \quad (10.3)$$

The coefficients are rational and are determined successively by

$$q_0 = 1, \quad (10.4)$$

$$q_j = [w^{j+1}] \{3\log(1 + w^2q(w)) - \log(1 - w + w^2q(w))\}, \quad (10.4)$$

$$r_j = 2q_j + [w^{j-2}]q(w)^2. \quad (10.5)$$

For every $J \geq 0$, truncation after $r_J t^{-J}$ has absolute error at most

$$\frac{129}{256c^2} \frac{(20/t)^{J+1}}{1 - 20/t}. \quad (10.6)$$

Proof. Divide (10.2) by w and rearrange as the fixed-point equation

$$q = \frac{2}{w} \log(1 + w^2 q) - \frac{1}{w} \log\left(1 - \frac{w}{1 + w^2 q}\right). \quad (10.7)$$

As before, the apparent singularities at zero are removable. For $|w| \leq r = 1/20$, $|q - 1| \leq 1/4$ and $Q = 5/4$, the right-hand side minus 1 has modulus at most

$$\frac{2rQ}{1 - r^2Q} + \frac{r^2Q}{1 - r^2Q} + \frac{r}{2(1 - r^2Q)(1 - r^2Q - r)} < \frac{1}{4}.$$

Here the last two terms bound the second logarithm in (10.7) minus 1. Differentiating the original rearrangement of (10.2) bounds the q -derivative by

$$\frac{3r}{1 - r^2Q} + \frac{r}{1 - r - r^2Q} < \frac{1}{4}.$$

Uniform analytic contraction proves existence and uniqueness. Coefficient comparison gives (10.4); its right side depends only on earlier q coefficients.

For real $w = 1/t > 0$, set $\mu = t + wq(w)$ and $x = a + \mu^2/c^2$. The relation $M(x) = H(u)$, with $u = t^2/c^2$, is equivalent to

$$(\mu - t) - 2 \log(\mu/t) + \log(1 - \mu^{-1}) = 0.$$

Substitute $\mu/t = 1 + w^2q$ to obtain (10.2). The uniqueness of G then gives (10.3). Finally,

$$|2(q - 1) + w^2q^2| \leq \frac{129}{256}$$

on $|w| = 1/20$, so the same Cauchy argument as before proves (10.6). $\square$

The first correction terms are

$$G(y) = F(y) + \delta + \frac{5}{c^2t} + \frac{29}{3c^2t^2} + \frac{83}{6c^2t^3} + \frac{559}{60c^2t^4} + \mathcal{O}(t^{-5}). \quad (10.8)$$

The third coefficient is $83/6$, whereas the bias coefficient in (9.4) is $113/6$. Interchanging them amounts to replacing the unknown index by its uncorrected inverse inside a higher-order term, and produces an incorrect expansion.

At actual partition targets, the corrected model is exponentially accurate rather than exact:

$$G(p(n)) - n = \mathcal{O}(\mu^3 e^{-\mu/2}). \quad (10.9)$$

To see this, combine (3.3) with the positive logarithmic derivative (10.1), whose reciprocal is $\mathcal{O}(\mu)$ on a fixed-width neighborhood of n . The same bracketing argument as in Theorem 5.1 applies for sufficiently large n . The bound in (10.9) tends to zero, so rounding $G(p(n))$ eventually recovers n . No sharp starting index for that separate corrected-model rule is asserted here.

11 Arithmetic exponential corrections

The coefficients $A_k(n)$ create periodic fluctuations that no fixed power series in $n^{-1/2}$ can capture. We now isolate them at the inverse level with a controlled remainder.

Table 1: Exact coefficients of the two convergent inverse constructions. Here w_j belongs to $B(1/\mu)$ and r_j to $G(y) - F(y)$.

j	w_j	r_j
1	5	5
2	29/3	29/3
3	113/6	83/6
4	823/20	559/60
5	1517/15	-611/30
6	338257/1260	-112459/1260
7	938657/1260	-101329/630
8	10713193/5040	-228677/5040
9	15593947/2520	1709167/2520
10	317064037/17325	325098619/138600
11	19540325693/356400	329128673/89100
12	1076961671111/6486480	-9006152791/6486480

Theorem 11.1 (Any fixed finite set of arithmetic sectors). *Fix an integer $K \geq 2$. Put $z = 1/\mu$ and $\nu = \mu - v(z)/\mu$. As $n \rightarrow \infty$ through integers,*

$$d_n = B(1/\mu) - \frac{2\nu^2}{c^2(\nu - 2)} \sum_{k=2}^K \frac{A_k(n)}{\sqrt{k}} \frac{1 - k/\mu}{1 - 1/\mu} e^{-(1-1/k)\mu} + \mathcal{O}_K(\mu^3 e^{-(1-1/(K+1))\mu}). \quad (11.1)$$

The implied constant can depend on the fixed integer K , but not on n or its residue class.

Proof. Splitting $J(\mu/k)$ into exponentials, the positive-exponential part of the k th Rademacher summand is

$$P_{k,+}(n) = \frac{A_k(n)}{4\sqrt{3}m\sqrt{k}} e^{\mu/k} (1 - k/\mu).$$

Its ratio to $M(n)$ is

$$E_k(n) = \frac{A_k(n)}{\sqrt{k}} \frac{1 - k/\mu}{1 - 1/\mu} e^{-(1-1/k)\mu}. \quad (11.2)$$

For fixed K , the negative-exponential parts of the first K terms, after division by $M(n)$, are $\mathcal{O}_K(e^{-(1+1/K)\mu})$. Lemma 3.1 bounds the remaining exact Rademacher tail divided by $M(n)$ by $\mathcal{O}_K(\mu^2 e^{-(1-1/(K+1))\mu})$. Therefore

$$\epsilon_n = \sum_{k=2}^K E_k(n) + \mathcal{O}_K(\mu^2 e^{-(1-1/(K+1))\mu}). \quad (11.3)$$

In particular, taking $K = 2$ first shows $\epsilon_n = \mathcal{O}(e^{-\mu/2})$: the polynomial factor in the next exponential is absorbed by the gap between its exponent and $1/2$.

Let $\Phi(\ell) = F(e^\ell)$. At $\ell = \log M(n)$,

$$\Phi'(\ell) = \frac{2\nu^2}{c^2(\nu - 2)} = \mathcal{O}(\mu). \quad (11.4)$$

On a bounded neighborhood of this argument, $\Phi''(\ell) = \mathcal{O}(1)$. Indeed $\ell = \nu - 2 \log \nu + \log(c^2/(4\sqrt{3}))$, so $d\nu/d\ell = \nu/(\nu - 2)$, and direct differentiation of (11.4) is bounded for large ν . Taylor expansion in $h = \log(1 + \epsilon_n)$ gives

$$F(p(n)) - F(M(n)) = \frac{2\nu^2}{c^2(\nu - 2)}\epsilon_n + \mathcal{O}(\mu\epsilon_n^2).$$

The nonlinear error is $\mathcal{O}(\mu e^{-\mu})$, smaller than the stated remainder for every fixed K . Substitution of (11.3) and the identity $d_n = B - (F(p(n)) - F(M(n)))$ proves the result. $\square$

Corollary 11.2 (Leading parity effect). *Define the analytic amplitude near $z = 0$ by*

$$\mathcal{A}(z) = \frac{(1 - z^2 v(z))^2 (1 - 2z)}{(1 - 2z - z^2 v(z))(1 - z)}, \quad \mathcal{A}(0) = 1. \quad (11.5)$$

Then

$$d_n = B(1/\mu) - (-1)^n \frac{\sqrt{2}\mu}{c^2} \mathcal{A}(1/\mu) e^{-\mu/2} + \mathcal{O}(\mu^3 e^{-2\mu/3}). \quad (11.6)$$

In particular the smooth-bias residual is eventually negative on even indices and positive on odd indices.

Proof. Take $K = 2$, use $A_2(n) = (-1)^n$, and substitute $\nu = \mu(1 - z^2 v)$. This simplifies the prefactor in (11.1) to the amplitude in (11.5). The quotient of the remainder by the leading exponential is $\mathcal{O}(\mu^2 e^{-\mu/6}) \rightarrow 0$, while $\mathcal{A}(1/\mu) \rightarrow 1$. $\square$

The first arithmetic layer explains why a smooth large- n expansion can miss small odd-index behavior. It does not by itself prove the finite exceptions, which were separately certified.

There is an important limitation to the word “transseries” here. The actions $1 - 1/k$ in (11.1) accumulate at 1, and nonlinear products of the first exponential layer also reach action 1. Theorem 11.1 is a rigorous *fixed finite-sector* statement below that accumulation point. It is not a proof that a naively ordered infinite formal transseries remains valid across the accumulation. Developing an appropriate resummation there is a genuine further question.

12 Consequences for differences and perturbations

12.1 Fixed-order eventual alternating differences

Write $\Delta f(n) = f(n+1) - f(n)$ and let $(1/2)_r = (1/2)(3/2) \cdots (r-1/2)$, with $(1/2)_0 = 1$.

Theorem 12.1 (Every fixed finite-difference order). *For every fixed integer $r \geq 0$,*

$$(-1)^r \Delta^r (d_n - \delta) \sim \frac{5}{c^3} (1/2)_r n^{-r-1/2} \quad (n \rightarrow \infty). \quad (12.1)$$

Consequently d_n is eventually strictly decreasing and strictly convex, and every fixed-order alternating difference is eventually positive.

Proof. Let $f(x) = B((c\sqrt{x-a})^{-1}) - \delta$. The convergent analytic series near infinity gives, for fixed r ,

$$f^{(r)}(x) = (-1)^r \frac{5}{c^3} (1/2)_r x^{-r-1/2} + \mathcal{O}_r(x^{-r-1}).$$

Termwise differentiation is justified by local uniform convergence on a complex neighborhood of each sufficiently large positive ray. For $r \geq 1$,

$$\Delta^r f(n) = \int_{[0,1]^r} f^{(r)}(n + t_1 + \cdots + t_r) dt_1 \cdots dt_r.$$

The case $r = 0$ is the original expansion. On the other hand, Theorem 5.1 bounds the discrete remainder $d_n - \delta - f(n)$ by an exponentially small function of $\sqrt{n}$. Its r th finite difference is bounded by the sum of 2^r such local bounds, which is smaller than every fixed negative power of n . Combining these facts proves (12.1). $\square$

The order of quantifiers matters: for each fixed r there exists a threshold N_r . No single threshold valid for all r is claimed. The small-index increase from d_{10} to d_{11} also shows why “eventually decreasing” cannot simply be strengthened to monotonicity throughout the OEIS conjecture’s range.

12.2 The natural noise scale for recovering an index

Proposition 12.2 (Inverse sensitivity at the critical scale). *Let $\eta \in \mathbb{R}$ be fixed and suppose*

$$y_n = p(n) \exp\left(\frac{\eta}{\sqrt{n}} + o(n^{-1/2})\right). \quad (12.2)$$

Then

$$F(y_n) - n \longrightarrow -\delta + \frac{2\eta}{c}. \quad (12.3)$$

In particular, nearest-integer recovery eventually succeeds if

$$\frac{c}{2}(\delta - \tfrac{1}{2}) < \eta < \frac{c}{2}(\delta + \tfrac{1}{2}). \quad (12.4)$$

It eventually fails if the limit in (12.3) lies outside $[-1/2, 1/2]$. For the bias-corrected inverse $F(y_n) + \delta$, the corresponding sufficient condition is $|\eta| < c/4$.

Proof. On the logarithmic target scale, the derivative in (11.4), now evaluated near $\log p(n)$, is $(2/c)\sqrt{n} + \mathcal{O}(1)$. The mean value theorem applied to (12.2) therefore gives $F(y_n) - F(p(n)) \rightarrow 2\eta/c$. Subtracting $d_n \rightarrow \delta$ proves (12.3). A limit strictly inside the rounding interval implies eventual success, and a limit strictly outside its closure implies eventual failure. $\square$

At either boundary of (12.4), the leading limit alone does not decide success; the next error term and the tie convention can matter. The proposition deliberately leaves these boundary cases unasserted.

12.3 Relation to the forward approximation in A050811

The sequence A050811 rounds $H(n)$ itself and records an asymptotic formula for its difference from $p(n)$ [3]. Our inverse bias is consistent with that forward error. Indeed, $F(p(n)) = n - d_n$ and $d_n \rightarrow \delta$ imply

$$\log \frac{H(n)}{p(n)} = \log H(n) - \log H(n - d_n) \sim \frac{c\delta}{2\sqrt{n}}.$$

Consequently

$$H(n) - p(n) \sim \left(\frac{1}{\pi} + \frac{\pi}{72}\right) \frac{e^{\pi\sqrt{2n/3}}}{4\sqrt{2}n^{3/2}}. \quad (12.5)$$

Rounding $H(n)$ changes its value by at most $1/2$, negligible on this scale, so the same leading equivalent applies to $A050811(n) - p(n)$. The leading equivalent in (12.5) is already present in that OEIS entry; it is included as a consistency check, not as a novel formula. In particular, a small and bounded inverse error is compatible with an unbounded forward error.

13 Reproducibility and numerical illustrations

13.1 Three deliberately separate forms of verification

The archive separates exact proof checks from illustrative numerical calculations.

Finite proof certificate. `code/certify.py` uses only the Python standard library, with integer and rational arithmetic. It verifies Lemma 6.1, independently recomputes 400 partition values, and encloses the sharp constants. Its output is `results/certificate.json`; the companion CSV lists all 1,204 interval comparisons with exact endpoints. The analytic large-index proof is not encoded in this program.

Exact coefficient calculation. `code/series.py` evaluates the triangular recurrences over the rational numbers. The supplied run computes order 12 and writes `results/coefficients.json`. Both defining logarithmic equations are also verified by exact substitution through degree 13. Its output is exact algebra, not a high-precision fit to partition data.

Numerical experiments. `code/experiments.py` uses `mpmath` with 130 decimal digits. It evaluates the Lambert inverse, the analytic fixed points, the parity residual and the staircase formula. The resulting `results/numerics.json` is explicitly labeled numerical, not an interval certificate. These experiments check implementation and illustrate scale; they are not substitutes for the analytic proofs.

From the archive root, the reproduction commands are

```
python code/certify.py --output results/certificate.json
python code/series.py --order 12 --output results/coefficients.json
python code/experiments.py --output results/numerics.json
pdflatex -halt-on-error article.tex
pdflatex -halt-on-error article.tex
```

Only the third command needs the optional `mpmath` dependency. The `Makefile` automates these targets. The certificate must be run without Python's assertion-removing optimization flag.

13.2 The small indices that decide the theorem

Table 2 gives rounded decimal illustrations. Their role is expository; all strict comparisons used in the proof were separately certified with rational intervals.

Table 2: Inverse bias and nearest-integer recovery.

n	$p(n)$	d_n (approximately)	Recovery
2	2	0.796615481265	fails
3	3	0.756104945769	fails
4	5	0.502652583555	fails
5	7	0.645462483764	fails
6	11	0.438137374848	succeeds
7	15	0.569463332725	fails
8	22	0.450004070600	succeeds
9	30	0.505033636323	fails
10	42	0.440766683161	succeeds
11	56	0.498487575397	succeeds
12	77	0.420669769623	succeeds
13	101	0.473632529547	succeeds
14	135	0.431618414918	succeeds
15	176	0.452838038452	succeeds

The values at 4 and 9 lie just above $1/2$; the value at 11 lies just below. An argument based only on a few leading terms at those indices would be unreliable.

13.3 Checking the parity term and the convergent tails

Let

$$R_n = d_n - B(1/\mu), \quad L_n = -(-1)^n \frac{\sqrt{2}\mu}{c^2} \mathcal{A}(1/\mu) e^{-\mu/2}.$$

The quotient R_n/L_n should tend to 1 by Corollary 11.2. The following rounded values use the full analytic fixed point in B and $\mathcal{A}$, not a truncated polynomial.

n	R_n (approximately)	R_n/L_n (approximately)
100	$-1.5380713277 \cdot 10^{-5}$	0.995255662344
400	$-8.1517489953 \cdot 10^{-11}$	0.999897873292
401	$+7.9066922485 \cdot 10^{-11}$	1.000195579199
1000	$-4.2960209400 \cdot 10^{-17}$	0.999999260234
1001	$+4.2118871198 \cdot 10^{-17}$	1.000001383952
2000	$-3.0616805313 \cdot 10^{-24}$	0.999999994825
2001	$+3.0188441869 \cdot 10^{-24}$	0.999999992109

The oscillation of the ratios about 1 reflects the next arithmetic sectors, not a failure of the leading parity prediction.

For the twelve-term analytic bias approximation, the observed absolute errors in $B(1/\mu)$ at $n = 400, 1000, 2000$ are approximately $4.81 \cdot 10^{-18}$, $1.22 \cdot 10^{-20}$ and $1.33 \cdot 10^{-22}$. These are much smaller than the deliberately conservative Cauchy bounds. At $n = 2000$, the algebraic truncation error after twelve terms is larger than the leading parity residual. This illustrates why an exponentially small

signal cannot automatically be read off after a fixed short power-series truncation, even at high arithmetic precision.

For the corrected inverse, the experiments evaluate the twelve-term version of (10.3) at the target $M(n)$, where the exact answer is n . The errors (exact value minus approximation) at those same indices are about $-2.46 \cdot 10^{-19}$, $-6.22 \cdot 10^{-22}$ and $-6.79 \cdot 10^{-24}$. This tests the *inverse* coefficient family independently of the forward-bias coefficients.

14 Further research questions and topics

The following questions are proposed continuations of the proved results. Except for the OEIS conjecture resolved above, no claim is made that every formulation has already appeared as an open problem in the literature.

1. Locate the nearest complex singularities. The radius $1/20$ is a convenient sufficient radius, not an optimal one. Determine the exact radii of convergence of v and q , the dominant singularities on their convergence circles, and the resulting large-order asymptotics of w_j and r_j . The fixed-point equations suggest studying simultaneous vanishing of the implicit equation and its derivative in the unknown function. A rigorous answer would give much sharper coefficient and truncation bounds than the elementary disk estimates.

2. Prove or refute positivity of every bias coefficient. All v_j are nonnegative by the triangular recurrence, and the first twelve w_j are positive. Determine whether $w_j > 0$ for every $j \geq 1$. If true, seek a positive recurrence or a combinatorial interpretation that explains the cancellation in $2v - z^2v^2$. Such a result could produce monotone lower approximants to the smooth bias. The sign-changing corrected-inverse coefficients r_j provide a useful contrast: positivity must not be inferred merely from invertibility.

3. Find sharp thresholds for finite differences. For fixed r , Theorem 12.1 proves eventual positivity of $(-1)^r \Delta^r(d_n - \delta)$. Determine the least threshold N_r , starting with monotonicity and convexity, and study the growth of N_r with r . The parity term suggests that a joint regime with increasing r is substantially different from the fixed- r theorem. An effective proof should track both differentiation of the analytic sector and the amplification of alternating arithmetic errors.

4. Understand the extremum on every terminal range. For an integer $N \geq 2$, define $\beta(N) = \sup_{n \geq N} d_n$. We determined it sharply for $N = 2$ and $N = 10$. Classify the maximizing index as a function of N , identify where that index changes, and derive an effective description of the optimal shift $(\beta(N) + \delta)/2$. Eventual monotonicity would imply $\beta(N) = d_N$ beyond a threshold, but does not describe the finite transition region.

5. Resolve the accumulation of exponential actions. The sector actions $1 - 1/k$ accumulate at 1, while the square of the parity correction has action 1. Construct a mathematically suitable resummation or grouped expansion through this level. The target should be a theorem with a uniform remainder when the number of retained Rademacher terms grows with n , not a formal rearrangement of infinitely many exponentials. The exact convergent Rademacher series supplies a natural starting point.

6. Sharpen inverse algorithms and their certification. Replace the simple two-candidate theorem by an adaptive method that combines the convergent correction series, interval Lambert evaluation and an effective Rademacher routine. Prove bit-complexity bounds in the size of the target, including inputs near the floor boundaries in (8.2). The one-comparison result alone does not resolve the cost of certifying that floor. Another concrete task is to determine the sharp threshold for rounding the corrected model $G(p(n))$.

7. Extend exact inverse rounding beyond ordinary partitions. For partitions into distinct parts, colored partitions, or restricted partition families with controlled exponential expansions, identify the correct inverse scaling, limiting bias and possible successful constant shifts. The appropriate analogue of the present theorem is a global rounding statement with explicit exceptional indices, not only an inverse asymptotic equivalent. Families whose forward relative error falls exactly at the inverse sensitivity scale are especially natural.

8. Analyze the critical perturbation boundaries. At $\eta = c(\delta \pm 1/2)/2$, Proposition 12.2 leaves a half-integer limiting displacement. Add the next perturbation term, for example θ/n , and classify eventual success or failure. After sufficiently many algebraic terms are canceled, the arithmetic parity term can become the first decision-making term. This offers a precise setting in which a beyond-all-orders correction controls an integer-valued outcome.

9. Extract residue-sensitive error bounds. The $k = 2$ term distinguishes even and odd indices. Incorporate the $k = 3, 4, \dots$ terms to obtain sharp inequalities on fixed residue classes and to reduce the analytic-to-finite cutoff. A useful deliverable would be a small rational certificate for the least possible cutoff in an explicit proof architecture, rather than merely a longer table of numerical confirmations.

10. Formalize the proof and expose its dependencies. Formalize the interval operations, the finite partition product and Lemma 6.1 in Lean or Rocq. Separately formalize the analytic contraction and inverse-error arguments. The large external dependency is Rademacher’s formula: a formal development should clearly distinguish assuming an already-established statement of that theorem from formalizing its proof. A verified certificate checker would reduce the trusted code for the finite step without disguising that analytic boundary.

15 Conclusion

The conjecture in A306631 is an exact-recovery phenomenon that can be proved from classical partition analysis, but it is not a consequence of the leading asymptotic equivalent alone. The bias is bounded away from zero, approaches $1/24 + 3/\pi^2$, and has small-index arithmetic fluctuations decisive for rounding. The combination of an explicit infinite-range estimate and an exact rational finite certificate resolves the conjecture and identifies its complete exception set.

The sharp extremum at 11 then determines an optimal uniform shift, a numerical safety margin, and a two-candidate staircase inversion. The convergent analytic corrections and the separate arithmetic exponential layers explain how these discrete facts sit inside a more precise inverse theory. These are the concrete mathematical outcomes of the report; neither a sweeping priority claim nor a general solution to partition-function inversion is required for them.

A A compact proof dependency ledger

Result or object	Dependency and status
Rademacher identity	Quoted classical theorem, in the normalization of Johansson [4]. Not proved by the code.
Elementary tail	Derived here from $ A_k(n) \leq k$ and an integral bound on $J(t)$.
Convergent bias model	Analytic contraction, exact algebra and Cauchy estimates, proved here.
All $n \geq 400$	Relative Rademacher error, derivative lower bound and an explicit bracketing argument, proved here.
$2 \leq n \leq 399$	Exhaustive strict comparisons with rational interval endpoints; standard-library certificate supplied and executed.
Partition data	Exact finite-product computation independently checks all values used by the finite certificate.
Sharp rounding and extrema	Analytic infinite range plus the exact finite lemma.
Optimal shifts and staircase	Consequences of the sharp bias extrema and monotonicity.
Inverse coefficient lists	Exact rational recurrences, not numerical fitting.
Arithmetic corrections	Exact Rademacher sectors and Taylor expansion of the logarithmic inverse; fixed finite number of sectors.
Numerical tables	High-precision illustrations; not used as proof of strict inequalities.
Priority and formal status	Unrefereed report; limited search; no Lean/Rocq kernel checking claimed.

B Suggested OEIS update and source audit

An editor-ready draft is included as `OEIS_update_draft.txt`. Its central replacement for the conjectural statement is the following mathematical content, subject to independent editorial review:

For every integer $n \geq 10$, $A306631(p(n)) = n$. For $n \geq 2$, the only exceptions to this identity are 2, 3, 4, 5, 7, 9, and at each exception the value is $n - 1$.

A second useful comment would state the ceiling identity and the sharp bias bounds, with a reference to a stable, publicly available version of this report after review. No OEIS submission was made as part of this work.

The source audit distinguished the entry's own revision marker from the site-wide footer date. The A306631 internal page inspected for this report displayed its formula and conjecture, with entry revision **#16, Feb 16 2025**. The report date records when that content was checked, not a claim that the conjecture was first posted on that date. The entry credits Jean-François Alcover and dates its origin to 2 March 2019 [1].

The limited bibliographic check did not locate a prior proof of this specific exact-rounding statement. That negative search result is not a proof of novelty. The mathematical assertions are therefore stated as theorems proved in this report, with classical inputs credited, without claiming that every reformulation or consequence is historically unprecedented.

References

- [1] OEIS Foundation Inc., *A306631: Inverse of the Hardy–Ramanujan asymptotic partition function*. Entry credited to Jean-François Alcover, 2 March 2019. Internal revision #16 displayed 16 February 2025; consulted 3 October 2026. <https://oeis.org/A306631/internal>.
- [2] OEIS Foundation Inc., *A000041: Number of partitions of n* . <https://oeis.org/A000041>.
- [3] OEIS Foundation Inc., *A050811: Partition numbers rounded to nearest integer given by the Hardy–Ramanujan approximate formula*. Consulted 3 October 2026. <https://oeis.org/A050811/internal>.
- [4] F. Johansson, *Efficient implementation of the Hardy–Ramanujan–Rademacher formula*, LMS Journal of Computation and Mathematics **15** (2012), 341–359. <https://doi.org/10.1112/S1461157012001088>. Preprint version 2: <https://arxiv.org/abs/1205.5991v2>.
- [5] K. Banerjee, P. Paule, C.-S. Radu and C. Schneider, *Error bounds for the asymptotic expansion of the partition function*, 2022, arXiv:2209.07887. <https://arxiv.org/abs/2209.07887>. Cited for context on forward error bounds, not as a hidden proof assumption for the finite cutoff used here.
- [6] ProveIt repository maintained by V. Reshetnikov, *All orders asymptotics and asymptotic inversion for A292507*, report dated 1 October 2026, with repository integration notes dated 2 October 2026. Inspected at commit d68b65ea064084fb121df9bb431b21d086f7be81, path SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-a-symptotics/a292507-binomial-partition-transform/article.tex. Repository: <https://github.com/VladimirReshetnikov/ProveIt>. This repository source is methodological context, not a proof dependency.